Two-Page Theorem Summary

Stationary Diffusion-Driven Instability in Classical Mass Action

External-audit version

13 August 2026

Decision problem. The input is a rationally encoded finite indexed reaction network $y_r \to y_r'$, with source matrix $Y = [y_r]$ and stoichiometric matrix $\Gamma = [y_r' - y_r]$. Write $S = \operatorname{im}\Gamma$. Decide whether there are strictly positive rate constants, a strictly positive homogeneous equilibrium, and strictly positive diagonal diffusion coefficients for all species such that the homogeneous Jacobian is Hurwitz on S and some nonzero Neumann mode has a strictly positive real eigenvalue.

Exact characterization. For $v > 0$, $\Gamma v = 0$, and $h > 0$, set

$$A(v) = \Gamma \operatorname{diag}(v) Y^T, \qquad J(v,h) = A(v) \operatorname{diag}(h).$$

Choose any rational basis B of S and rational left inverse C, and set $R(v,h) = CJ(v,h)B$. The answer is YES exactly when there are v, h and a nonempty principal set I such that

$$\Gamma v = 0, \qquad R(v,h) \text{ is Hurwitz}, \qquad (-1)^{|I|} \det A(v)[I,I] < 0.$$

The converse mass-action realization is $x_i^* = h_i^{-1}$ and $k_r = v_r/(x^*)^{y_r}$.

The fixed-matrix step is

$$\exists D \succ 0 : \ J - D \text{ has a positive real eigenvalue} \iff \exists I : \ (-1)^{|I|} \det J[I,I] < 0.$$

It follows from the principal-minor expansion of $\det(D - J)$ and a multiscale assignment that makes one negative coefficient uniquely dominant. Column scaling gives

$$(-1)^{|I|} \det J[I,I] = \Big(\prod_{i \in I} h_i\Big)(-1)^{|I|} \det A(v)[I,I].$$

Conservation. If $J(\mathbb{R}^n) \subseteq S$ and $J|_S$ is Hurwitz, then $\mathbb{R}^n = S \oplus \ker J$ and the conservation zero eigenvalues are semisimple. A nonzero Neumann eigenfunction has zero spatial mean, so its species amplitude is unrestricted; one must test $J - q^2 D$ on the full species space.

Complexity. A Boolean selector $b_i^2 - b_i = 0$, with $S_b = \mathrm{diag}(b)$, satisfies

$$\det(I - S_b - S_b A(v) S_b) = (-1)^{|I|} \det A(v)[I, I], \qquad I = \{i : b_i = 1\}.$$

Determinant and Hurwitz circuits have polynomial size, so the decision problem belongs to the existential theory of the reals.

The problem is NP-hard under a polynomial-time many-one reduction from Partition. For $m = k^2 > \ell$, the source family is

$$Q = I + aa^T, \quad \beta = 1 - \frac{1}{2m(1 + a^T a)}, \quad B(x, y) = \begin{pmatrix} -kQ & y \\ x^T & k\beta \end{pmatrix}.$$

Its open cube contains a Hurwitz matrix exactly for YES partition instances. A row-splitting lift $L(q)$ is similar to $\left(\begin{smallmatrix} B(q) & U \\ 0 & -I \end{smallmatrix}\right)$. A separate determinant argument proves

$$\exists q, D \succ 0 : \; L(q)D \text{ Hurwitz} \iff \text{Partition} = \text{YES},$$

so arbitrary equilibrium/right scaling cannot create a false YES. An exact row-segment reaction construction realizes the full family $\{RL(q) : R \succ 0\}$ with independent steady-flux constraints. The network has $3m + 1$ species, $8m + 2$ reactions, full stoichiometric rank, and columns $\pm e_i$. Its complex molecularity is unbounded, and the hardness is weak.

Fixed species count. Let $K = \{v \geq 0 : \Gamma v = 0\}$ and $P = \{\Gamma \,\mathrm{diag}(v) Y^T : v \in K\}$. If a positive flux exists, positive fluxes project exactly to $\mathrm{ri}\, P$. Every extreme ray of K is a positive circuit supported on at most $\mathrm{rank}\, \Gamma + 1 \leq n + 1$ reactions. For fixed n these rays, the fixed-dimensional projected cone, and a relative-facet description are computable in polynomial time. The remaining real formula has a fixed number of variables, giving polynomial bit complexity for every fixed n.

Certificates and limitations. YES instances have finite real-algebraic sample certificates; NO instances have finite Real-Nullstellensatz identities. No polynomial size or practical general generator is claimed. The theorem concerns weak stationary linear instability only: it does not require a first, simple, or transverse crossing, does not exclude an earlier wave instability, does not prove a nonlinear patterned state, and does not give a finite motif characterization.